WEEK-1

Part A – NumPy

Question 1:
Create a NumPy array containing the integers from 1 to 20.
Perform the following operations:
1. Display the array.
2. Find the shape of the array.
3. Find the maximum and minimum values.
4. Calculate the mean and standard deviation.
5. Reshape the array into a 4 × 5 matrix.
6. Display the second row of the matrix.
7. Display the third column of the matrix.

Code:
```
import numpy as np
print("1.Display the array")
arr=np.arange(1,21)
print(arr)
print("\n 2.Find the Shape of the array",arr.shape)
print(f"\n 3.Find the Maximum values and Minimum values:{arr.max()},{arr.min()}")
print(f"\n 4.Find the Mean and Standard Deviation:{arr.mean()},{arr.std()}")
matrix=arr.reshape(4,5)
print("\n 5.Reshape the array into 4 x 5 matrix:",matrix)
print("\n 6.Display the second row of the matrix:",matrix[1,:])
print("\n 7.Display the third column of the matrix:”,matrix[:,2])
```

Output:

```
(.venv) mohanatumula@Mohanas-MacBook-Air Machinelearning % /Users/mohanatumula/Machinelearning
v/bin/python /Users/mohanatumula/Machinelearning/question1.py
1.Display the array
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20]

 2.Find the Shape of the array (20,)

 3.Find the Maximum values and Minimum values:20,1

 4.Find the Mean and Standard Deviation:10.5,5.766281297335398

 5.Reshape the array into 4 x 5 matrix: [[ 1  2  3  4  5]
 [ 6  7  8  9 10]
 [11 12 13 14 15]
 [16 17 18 19 20]]

 6.Display the second row of the matrix: [ 6  7  8  9 10]

 7.Display the third column of the matrix: [ 3  8 13 18]
```

Question 2:
Create two NumPy arrays:
A = [10, 20, 30, 40, 50]
B = [5, 10, 15, 20, 25]
Perform:
1. Addition of A and B.
2. Subtraction of B from A.
3. Element-wise multiplication.
4. Element-wise division.
5. Find the sum of all elements in A.

Code:

```
import numpy as np
A=np.arange(10,51,10)
print("\n Array A:",A)
B=np.arange(5,26,5)
print("\n Array B:",B)
print("\n Addition of A and B:",A+B)
print("\n Subtraction of B from A:",A-B)
print("\n Element-wise multiplication:",A*B)
print("\n Element-wise division:",A/B)
print("\n Find the sum of all elements in A",A.sum())
```

Output:

```
(.venv) mohanatumula@Mohanas-MacBook-Air Machinelearning % /Users/mohanatumul
a/Machinelearning/.venv/bin/python /Users/mohanatumula/Machinelearning/lab1qu
estion2.py

 Array A: [10 20 30 40 50]

 Array B: [ 5 10 15 20 25]

 Addition of A and B: [15 30 45 60 75]

 Subtraction of B from A: [ 5 10 15 20 25]

 Element-wise multiplication: [  50  200  450  800 1250]

 Element-wise division: [2. 2. 2. 2. 2.]

 Find the sum of all elements in A 150
```

Part B – Pandas

Question 3:
Create a DataFrame using the following student data:

Roll No	Name	Marks
101	Ravi	85
102	Priya	92
103	Kiran	78
104	Anu	88
105	Sita	95

Perform the following operations:
1. Display the DataFrame.
2. Display the first three records.
3. Find the average marks.
4. Find the student with the highest marks.
5. Find the student with the lowest marks.
6. Add a new column called "Grade" using:
o A for Marks ≥ 90
o B for Marks between 80 and 89
o C for Marks < 80

Code:

```
import pandas as pd
data={
    "roll no":[101,102,103,104,105],
    "Name":["Ravi","Priya","Kiran","Anu","Sita"],
    "Marks":[85,92,78,88,95]
}
df=pd.DataFrame(data)
print("1.Student Data Frame")
print(df)
print("2.First Three Records")
print(df.head(3))
averagemark=df["Marks"].mean()
print("3.Average Marks")
print(f"average:{averagemark}")
print("4.Find the Student with the maximum marks:")
maxmarkstudent=df.loc[df["Marks"].idxmax(),"Name"]
print(f"{maxmarkstudent}")
print("5.Find the Student with low marks:")
minmarksstudent=df.loc[df["Marks"].idxmin(),"Name"]
print(f"{minmarksstudent}")
df["Grade"]=None
df.loc[df["Marks"]>=90,"Grade"]="A"
```

df.loc[(df["Marks"]>80)&(df["Marks"]<90),"Grade"]="B"
df.loc[df["Marks"]<80,"Grade"]="C"
print("6.Table After add a new column grade")
print(df)

Output:

```
(.venv) mohanatumula@Mohanas-MacBook-Air Machinelearning % /Users/mohanatumul
a/Machinelearning/.venv/bin/python /Users/mohanatumula/Machinelearning/lab1qu
estion3.py
1.Student Data Frame
   roll no   Name  Marks
0      101   Ravi     85
1      102  Priya     92
2      103  Kiran     78
3      104    Anu     88
4      105   Sita     95
2.First Three Records
   roll no   Name  Marks
0      101   Ravi     85
1      102  Priya     92
2      103  Kiran     78
3.Average Marks
average:87.6
4.Find the Student with the maximum marks:
Sita
5.Find the Student with low marks:
Kiran
6.Table After add a new column grade
   roll no   Name  Marks Grade
0      101   Ravi     85     B
1      102  Priya     92     A
2      103  Kiran     78     C
3      104    Anu     88     B
4      105   Sita     95     A
```

Question 4

Create a DataFrame containing the following information:

Product	Price	Quantity
Laptop	50000	10
Mouse	500	50
Keyboard	1200	30
Monitor	10000	15

Perform:

1. Calculate the total value of each product using:
Total Value = Price × Quantity
2. Add the Total Value column.
3. Find the product with the highest total value.
4. Calculate the overall inventory value.

Code:

```
import pandas as pd
data={
    "product":("Laptop","Mouse","Keyboard","Monitor"),
    "price":(50000,500,1200,10000),
    "Quantity":(10,50,30,15)
}
df=pd.DataFrame(data)
print(df)
print("1 &2.Calculate and add the Total value column to table")
df["Totalvalue"]=df["price"]*df["Quantity"]
print(df)
print("3.Find the Product with the highest total value")
Highpurchasedproduct=df.loc[df["Totalvalue"].idxmax(),"product"]
print(f"The product with the highest total value:{Highpurchasedproduct}")
print("4.Calculate the overall inventory value")
overall_inventory_value=df["Totalvalue"].sum()
print(f' overall inventory value:{overall_inventory_value}")
```

Output:

```
(.venv) mohanatumula@Mohanas-MacBook-Air Machinelearning % /Users/mohanatumul
a/Machinelearning/.venv/bin/python /Users/mohanatumula/Machinelearning/lab1qu
estion4.py
    product  price  Quantity
0    Laptop  50000        10
1     Mouse    500        50
2  Keyboard   1200        30
3   Monitor  10000        15
1 &2.Calculate and add the Total value column to table
    product  price  Quantity  Totalvalue
0    Laptop  50000        10      500000
1     Mouse    500        50       25000
2  Keyboard   1200        30       36000
3   Monitor  10000        15      150000
3.Find the Product with the highest total value
The product with the highest total value:Laptop
4.Calculate the overall inventory value
overall inventory value:711000
```

Part C – Dataset Exploration

Question 5

Download the Iris dataset from Scikit-learn and perform the following:
1. Load the dataset into a Pandas DataFrame.
2. Display the first five records.
3. Find the number of rows and columns.
4. Display column names.
5. Check for missing values.
6. Generate summary statistics using describe().
7. Find the number of samples belonging to each class

Code:

```
import pandas as pd
from sklearn.datasets import load_iris
print("Load the dataset into a pandas DataFrames")
iris=load_iris(as_frame=True)
df=iris.frame
print("2.First 5 records")
print(df.head(5))
print("\n 3.shape of Dataset")
print(f"Rows:{df.shape[0]},columns:{df.shape[1]}")
print("4.column names ")
print(df.columns.tolist())
print("5.Missing values count")
print(df.isnull().sum())
print("6.Summary Statistics")
print(df.describe())
print("7.class Distribution")
print(df['target'].value_counts())
```

Output:

```
/.venv/bin/python /Users/mohanatumula/Machinelearning/lab1question5.py
2.First 5 records
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  target
0                5.1               3.5                1.4               0.2       0
1                4.9               3.0                1.4               0.2       0
2                4.7               3.2                1.3               0.2       0
3                4.6               3.1                1.5               0.2       0
4                5.0               3.6                1.4               0.2       0

 3.shape of Dataset
Rows:150,columns:5
4.column names
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)', 'target']
5.Missing values count
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
target               0
dtype: int64
6.Summary Statistics
       sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)      target
count         150.000000        150.000000         150.000000        150.000000  150.000000
mean            5.843333          3.057333           3.758000          1.199333    1.000000
std             0.828066          0.435866           1.765298          0.762238    0.819232
min             4.300000          2.000000           1.000000          0.100000    0.000000
25%             5.100000          2.800000           1.600000          0.300000    0.000000
50%             5.800000          3.000000           4.350000          1.300000    1.000000
75%             6.400000          3.300000           5.100000          1.800000    2.000000
max             7.900000          4.400000           6.900000          2.500000    2.000000
7.class Distribution
target
0    50
1    50
2    50
Name: count, dtype: int64
```